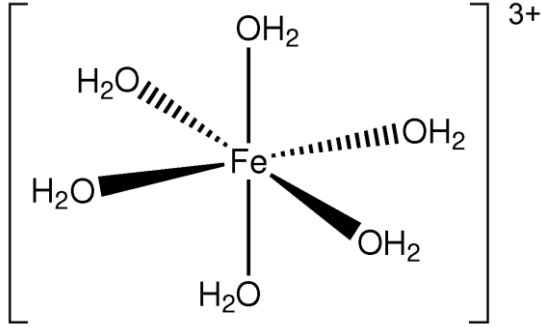


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)			$1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 (4s^0)$ (1) partially-filled d-orbitals (1)	1	1		2		
	(b)			award (1) for either of following the energy of the (4s and) 3d-orbitals are all similar the ionisation energies to remove the (4s and) 3d-electrons are similar	1			1		
	(c)			 must show clear octahedral structure with bonds between Fe and O atoms in water	1			1		
	(d)	(i)		different ligands cause different amount of d-orbital splitting (1) so different frequencies/wavelengths of light are absorbed (and different frequencies/wavelengths are transmitted/reflected) (1)	1	1		2		
		(ii)		find a wavelength absorbed by $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ but not by $[\text{Fe}(\text{H}_2\text{O})_5(\text{OH})]^{2+}$ / any other species in the mixture		1		1		1